```
/* ============================================================
   COMPLETE ARIMA INTERVENTION ANALYSIS - CORRECTED VERSION
   4 Candidate Models + Box Plot + Pulse Analysis + 24-Month Forecast

   Data: Sampah Plastik Kota Palu (Januari 2015 - Agustus 2025)
   Intervention Date: March 2022
   Forecast Period: 24 months (September 2025 - August 2027)

   Author: Jeremy Filippo G G (5003201183)
   Date: January 2026
   ============================================================ */

options nodate nonumber linesize=150;

/* ============================================================
   PART 0A: DATA PREPARATION
   ============================================================ */

proc import datafile="/home/u63802153/sasuser.v94/Hasil SAS/Data Sampah Plastik Kota Palu 2015-2025.xlsx"
    out=plastik_raw
    dbms=xlsx
    replace;
    getnames=yes;
    datarow=2;
run;

title "DATA STRUCTURE VERIFICATION";
proc contents data=plastik_raw varnum; run;

title "FIRST 10 ROWS";
proc print data=plastik_raw(obs=10); run;

data plastik;
    set plastik_raw;

    array allnums[*] _NUMERIC_;
    _var_count = 0;

    do _idx = 1 to dim(allnums);
        if not missing(allnums[_idx]) then do;
            _var_count + 1;

            if _var_count = 1 and missing(Date) then
                Date = allnums[_idx];

            if _var_count = 2 and missing(Volume_Sampah) then
                Volume_Sampah = allnums[_idx];
        end;
    end;

    /* Create intervention variable - Policy starts March 2022 */
    if Date >= '01MAR2022'd then Intervention = 1;
    else Intervention = 0;

    format Date yymmdd10.;
    format Volume_Sampah comma12.2;

    if not missing(Date) and not missing(Volume_Sampah);

    keep Date Volume_Sampah Intervention;
run;

proc sort data=plastik; by Date; run;

title "DATA VERIFICATION AFTER PREPARATION";
proc print data=plastik(obs=20); run;

title "LAST 10 ROWS";
proc print data=plastik(firstobs=118); run;

title "SUMMARY STATISTICS";
proc means data=plastik n mean std min max median;
```

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    var Volume_Sampah;
run;

title "INTERVENTION DISTRIBUTION";
proc freq data=plastik;
    tables Intervention;
run;

/* ============================================================
   PART 0B: TIME SERIES VISUALIZATION WITH INTERVENTION
   ============================================================ */

title "==========================================================";
title2 "TIME SERIES PLOT WITH INTERVENTION";
title3 "==========================================================";

ods graphics on / reset imagename="timeseries_with_intervention" width=12in height=8in;

proc sgplot data=plastik;
    /* Plot the time series */
    series x=Date y=Volume_Sampah /
        lineattrs=(color=darkblue thickness=2 pattern=solid)
        markers
        markerattrs=(symbol=circlefilled size=8 color=darkblue)
        legendlabel="Plastic Waste Volume";

    /* Add vertical line for intervention */
    refline '01MAR2022'd / axis=x
        lineattrs=(color=red pattern=shortdash thickness=3)
        label="Policy Implementation: March 2022"
        labelloc=inside
        labelattrs=(size=10pt weight=bold color=red);

    /* Axis settings */
    xaxis label="Time (Monthly)"
          grid
          gridattrs=(color=lightgray pattern=dot)
          values=('01JAN2015'd to '01AUG2025'd by year)
          valuesformat=year4.;

    yaxis label="Plastic Waste Volume (tons)"
          grid
          gridattrs=(color=lightgray pattern=dot)
          min=0;

    /* Title */
    title height=14pt bold "Time Series of Plastic Waste Volume";
    title2 height=12pt "Kota Palu (January 2015 - August 2025)";
    title3 height=10pt italic "Red Line Indicates Policy Implementation in March 2022";
run;

ods graphics off;

/* Alternative: Split Plot (Before vs After) */
title "TIME SERIES COMPARISON: Before vs After Policy Implementation";

ods graphics on / reset imagename="timeseries_before_after" width=12in height=8in;

proc sgplot data=plastik;
    /* Different colors for before/after */
    styleattrs
        datacontrastcolors=(darkblue darkred)
        datalinepatterns=(solid solid);

    /* Plot series with GROUP by Intervention */
    series x=Date y=Volume_Sampah /
        group=Intervention
        lineattrs=(thickness=2)
        markers
        markerattrs=(size=6);

    /* Intervention line */
```

```
    refline '01MAR2022'd / axis=x
        lineattrs=(color=black pattern=dash thickness=2)
        label="Policy Implementation"
        labelloc=inside
        labelattrs=(size=10pt weight=bold);

    /* Axis */
    xaxis label="Time (Monthly)" grid;
    yaxis label="Plastic Waste Volume (tons)" grid;

    /* Legend */
    keylegend / title="Period:"
                location=inside
                position=topright
                across=1;

    /* Title */
    title height=14pt bold "Plastic Waste Volume: Pre vs Post Policy";
    title2 height=11pt "Blue = Before Policy (2015-Feb 2022) | Red = After Policy (Mar 2022-Aug 2025)";
run;

ods graphics off;

/* Statistical Summary: Before vs After */
title "DESCRIPTIVE STATISTICS: Before vs After Policy Implementation";

proc means data=plastik n mean std min max median;
    class Intervention;
    var Volume_Sampah;
    label Intervention = "Policy Period (0=Pre, 1=Post)";
run;

proc ttest data=plastik;
    class Intervention;
    var Volume_Sampah;
    title "T-TEST: Comparing Mean Volume Before vs After Policy";
run;

/* ============================================================
   PART 0C: BOX PLOT ANALYSIS BY YEAR
   ============================================================ */

title "==========================================================";
title2 "BOX PLOT ANALYSIS";
title3 "==========================================================";

/* Create Year variable for grouping */
data plastik_yearly;
    set plastik;
    Year = year(Date);
    Period = put(Year, 4.);

    /* Create intervention period label */
    if Intervention = 0 then Period_Label = "Pre-Policy";
    else Period_Label = "Post-Policy";
run;

/* Box Plot: Annual Distribution */
title "BOX PLOT: Annual Distribution of Plastic Waste Volume";

ods graphics on / reset imagename="boxplot_annual" width=9in height=5in;

proc sgplot data=plastik_yearly;
    vbox Volume_Sampah / category=Period
                         fillattrs=(color=lightblue)
                         outlierattrs=(color=red symbol=circlefilled size=8)
                         whiskerattrs=(thickness=2)
                         boxwidth=0.6;

    /* Add reference line for intervention year */
    refline "2022" / axis=x
        lineattrs=(color=red pattern=dash thickness=2)
```

```
        label="Policy Year"
        labelloc=inside
        labelattrs=(size=10pt weight=bold color=red);

    xaxis label="Year"
          labelattrs=(size=11pt weight=bold);
    yaxis label="Plastic Waste Volume (tons)"
          grid
          labelattrs=(size=11pt weight=bold);

    title height=14pt bold "Annual Box Plot of Plastic Waste Volume";
    title2 height=11pt "Distribution Comparison Across Years (2015-2025)";
run;

ods graphics off;

/* Box Plot: Pre vs Post Policy Comparison */
title "BOX PLOT COMPARISON: Before vs After Policy Implementation";

ods graphics on / reset imagename="boxplot_comparison" width=6in height=5in;

proc sgplot data=plastik_yearly;
    vbox Volume_Sampah / category=Period_Label
                         fillattrs=(color=lightblue transparency=0.3)
                         meanattrs=(symbol=diamondfilled size=12 color=red)
                         outlierattrs=(color=darkred symbol=circlefilled size=8);

    xaxis label="Policy Period"
          labelattrs=(size=12pt weight=bold);
    yaxis label="Plastic Waste Volume (tons)"
          grid
          labelattrs=(size=12pt weight=bold);

    title height=14pt bold "Box Plot: Pre-Policy vs Post-Policy Period";
    title2 height=11pt "Red Diamond = Mean | Box = IQR | Whiskers = Min/Max";
run;

ods graphics off;

/* Statistical Summary by Year */
title "ANNUAL STATISTICS SUMMARY";
proc means data=plastik_yearly n mean std min q1 median q3 max;
    class Year;
    var Volume_Sampah;
run;

/* ANOVA Test: Check if years are significantly different */
title "ANOVA: Testing Year Effect on Plastic Waste Volume";
proc anova data=plastik_yearly;
    class Year;
    model Volume_Sampah = Year;
    means Year / tukey;
run;
quit;

/* ============================================================
   PART 1: STATIONARITY & IDENTIFICATION
   ============================================================ */

title1 "==========================================================";
title2 "PART 1: STATIONARITY TESTS & ACF/PACF";
title3 "==========================================================";

title4 "1.1 ADF Test: Level Data";
proc arima data=plastik;
    identify var=Volume_Sampah stationarity=(adf=(1,2,3));
run;

title4 "1.2 ADF Test: First Difference";
proc arima data=plastik;
    identify var=Volume_Sampah(1) stationarity=(adf=(1,2,3));
run;
```

```
ods graphics on / reset imagename="acf_pacf_diff1" width=10in height=8in;
title4 "1.3 ACF & PACF: First Differenced Data";
proc arima data=plastik plots=all;
    identify var=Volume_Sampah(1) nlag=36;
run;
ods graphics off;

/* ============================================================
   PART 2: ESTIMATE 4 CANDIDATE MODELS (NO INTERVENTION)
   ============================================================ */

title1 "============================================================";
title2 "PART 2: 4 CANDIDATE MODELS ESTIMATION";
title3 "============================================================";

/* ------------------------------------------------------
   MODEL 1: ARIMA(0,1,1)(1,0,0)12
   - Regular: p=0, d=1, q=1
   - Seasonal: P=1, D=0, Q=0, s=12
   ------------------------------------------------------ */
title4 "MODEL 1: ARIMA(0,1,1)(1,0,0)12";
proc arima data=plastik plots(unpack)=all;
    identify var=Volume_Sampah(1) nlag=36;
    estimate q=1 p=(12) method=ml noconstant
             outest=m1_params outstat=m1_stats;
    forecast lead=0 out=m1_forecast;
run;

title5 "MODEL 1: Residual Diagnostics";
proc arima data=m1_forecast plots=all;
    identify var=residual nlag=24;
run;

/* ------------------------------------------------------
   MODEL 2: ARIMA(0,1,1)(2,0,0)12
   - Regular: p=0, d=1, q=1
   - Seasonal: P=2, D=0, Q=0, s=12
   ------------------------------------------------------ */
title4 "MODEL 2: ARIMA(0,1,1)(2,0,0)12";
proc arima data=plastik plots(unpack)=all;
    identify var=Volume_Sampah(1) nlag=36;
    estimate q=1 p=(12,24) method=ml noconstant
             outest=m2_params outstat=m2_stats;
    forecast lead=0 out=m2_forecast;
run;

title5 "MODEL 2: Residual Diagnostics";
proc arima data=m2_forecast plots=all;
    identify var=residual nlag=24;
run;

/* ------------------------------------------------------
   MODEL 3: ARIMA(0,1,1)(1,1,1)12
   - Regular: p=0, d=1, q=1
   - Seasonal: P=1, D=1, Q=1, s=12
   ------------------------------------------------------ */
title4 "MODEL 3: ARIMA(0,1,1)(1,1,1)12";
proc arima data=plastik plots(unpack)=all;
    identify var=Volume_Sampah(1,12) nlag=36;
    estimate q=(1)(12) p=(12) method=ml noconstant
             outest=m3_params outstat=m3_stats;
    forecast lead=0 out=m3_forecast;
run;

title5 "MODEL 3: Residual Diagnostics";
proc arima data=m3_forecast plots=all;
    identify var=residual nlag=24;
run;

/* ------------------------------------------------------
   MODEL 4: ARIMA(0,1,[1,6])(2,0,0)12
```

```
   - Regular: p=0, d=1, q=[1,6]
   - Seasonal: P=2, D=0, Q=0, s=12
   ------------------------------------------------------ */
title4 "MODEL 4: ARIMA(0,1,[1,6])(2,0,0)12";
proc arima data=plastik plots(unpack)=all;
    identify var=Volume_Sampah(1) nlag=36;
    estimate q=(1,6) p=(12,24) method=cls noconstant
             outest=m4_params outstat=m4_stats;
    forecast lead=0 out=m4_forecast;
run;

title5 "MODEL 4: Residual Diagnostics";
proc arima data=m4_forecast plots=all;
    identify var=residual nlag=24;
run;

/* ============================================================
   PART 3: MODEL COMPARISON - AIC & SBC
   ============================================================ */

title1 "==========================================================";
title2 "PART 3: MODEL COMPARISON (AIC & SBC)";
title3 "==========================================================";

data model_comparison;
    length Model $45;
    set m1_stats (in=a)
        m2_stats (in=b)
        m3_stats (in=c)
        m4_stats (in=d);
    where _STAT_ in ('AIC', 'SBC');

    /* CONSISTENT MODEL NAMING */
    if a then Model = "Model 1: ARIMA(0,1,1)(1,0,0)12";
    else if b then Model = "Model 2: ARIMA(0,1,1)(2,0,0)12";
    else if c then Model = "Model 3: ARIMA(0,1,1)(1,1,1)12";
    else if d then Model = "Model 4: ARIMA(0,1,[1,6])(2,0,0)12";
run;

proc transpose data=model_comparison out=comparison_wide;
    by Model;
    id _STAT_;
    var _VALUE_;
run;

proc sort data=comparison_wide; by AIC; run;

title4 "MODEL RANKING (Lower AIC = Better)";
proc print data=comparison_wide noobs;
    format AIC SBC 12.2;
run;

/* Calculate AIC differences */
data comparison_delta;
    set comparison_wide;

    if _N_ = 1 then do;
        Best_AIC = AIC;
        Best_SBC = SBC;
        retain Best_AIC Best_SBC;
    end;

    Delta_AIC = AIC - Best_AIC;
    Delta_SBC = SBC - Best_SBC;

    if Delta_AIC < 2 then Evidence = "Substantial support";
    else if Delta_AIC < 4 then Evidence = "Weak support";
    else if Delta_AIC < 7 then Evidence = "Little support";
    else Evidence = "No support";
run;

title4 "MODEL COMPARISON WITH EVIDENCE WEIGHT";
```

```
proc print data=comparison_delta noobs;
    format AIC SBC Delta_AIC Delta_SBC 12.2;
    var Model AIC SBC Delta_AIC Delta_SBC Evidence;
run;

/* ============================================================
   PART 4: PARAMETER SIGNIFICANCE - ALL 4 MODELS
   ============================================================ */

title1 "==========================================================";
title2 "PART 4: PARAMETER SIGNIFICANCE CHECK";
title3 "==========================================================";

/* Macro to check parameter significance */
%macro check_params(model_num, model_name, df_params);

    proc transpose data=m&model_num._params out=temp_check;
        var _NUMERIC_;
        id _TYPE_;
    run;

    data temp_test;
        set temp_check;
        where EST ne . and STD ne .;

        t_value = EST / STD;
        df = 127 - &df_params;
        p_value = 2 * (1 - probt(abs(t_value), df));

        if p_value < 0.001 then Sig = "***";
        else if p_value < 0.01 then Sig = "**";
        else if p_value < 0.05 then Sig = "*";
        else Sig = "ns";

        keep _NAME_ EST STD t_value p_value Sig;
    run;

    title4 "&model_name - Parameter Estimates";
    proc print data=temp_test noobs;
        format EST STD 12.6 t_value 10.4 p_value pvalue6.4;
    run;

%mend;

/* CONSISTENT MODEL NAMING IN MACRO CALLS */
%check_params(1, MODEL 1: ARIMA(0,1,1)(1,0,0)12, 2);
%check_params(2, MODEL 2: ARIMA(0,1,1)(2,0,0)12, 3);
%check_params(3, MODEL 3: ARIMA(0,1,1)(1,1,1)12, 3);
%check_params(4, MODEL 4: ARIMA(0,1,[1,6])(2,0,0)12, 4);

/* ============================================================
   PART 5: SELECT BEST MODEL
   ============================================================ */

title1 "==========================================================";
title2 "PART 5: BEST MODEL SELECTION";
title3 "==========================================================";

/* Get the best model name */
data _null_;
    set comparison_wide(obs=1);
    call symputx('best_model', Model);
    call symputx('best_aic', put(AIC, 12.2));
    call symputx('best_sbc', put(SBC, 12.2));
run;

title4 "SELECTED BEST MODEL";
data best_model_info;
    length Info $100 Value $50;
    Info = "Best Model"; Value = "&best_model";
    output;
    Info = "AIC"; Value = "&best_aic";
```

```
    output;
    Info = "SBC"; Value = "&best_sbc";
    output;
run;

proc print data=best_model_info noobs;
run;

/* ============================================================
   PART 6: CROSS-CORRELATION WITH INTERVENTION
   ============================================================ */

title1 "==========================================================";
title2 "PART 6: CROSS-CORRELATION ANALYSIS";
title3 "==========================================================";

ods graphics on / reset imagename="crosscorr" width=8in height=6in;
title4 "Cross-Correlation: Volume vs Intervention";
proc arima data=plastik plots=all;
    identify var=Volume_Sampah(1)
             crosscorr=(Intervention(1))
             nlag=36;
run;
ods graphics off;

/* ============================================================
   PART 7: STEP INTERVENTION MODEL
   ============================================================ */

title1 "==========================================================";
title2 "PART 7: STEP INTERVENTION ANALYSIS";
title3 "==========================================================";

title4 "STEP MODEL: Permanent Effect (March 2022)";

proc arima data=plastik plots(unpack)=all;
    identify var=Volume_Sampah(1)
             crosscorr=(Intervention(1))
             nlag=36;

    estimate q=(1,6) p=(12,24)
             input=(0$(11,12)Intervention)
             method=cls
             noconstant
             outest=final_params
             outstat=final_stats;

    forecast lead=24 id=Date interval=month out=final_forecast;
run;

/* ============================================================
   PART 7A: PULSE INTERVENTION ANALYSIS
   ============================================================ */

title1 "==========================================================";
title2 "PART 7A: PULSE INTERVENTION ANALYSIS";
title3 "==========================================================";

/* Create Pulse Intervention Variables */
data plastik_pulse;
    set plastik;

    /* Step Intervention (existing) - Permanent effect */
    if Date >= '01MAR2022'd then Step_Intervention = 1;
    else Step_Intervention = 0;

    /* Pulse Intervention (only March 2022) - Temporary shock */
    if Date = '01MAR2022'd then Pulse_Intervention = 1;
    else Pulse_Intervention = 0;

    /* Gradual Intervention (linear ramp after policy) */
    if Date < '01MAR2022'd then Gradual_Intervention = 0;
```

```
    else Gradual_Intervention = intck('month', '01MAR2022'd, Date) + 1;
run;

/* Verify intervention variables */
title4 "INTERVENTION VARIABLES VERIFICATION";
proc print data=plastik_pulse(obs=100);
    where Date >= '01JAN2022'd;
    var Date Volume_Sampah Step_Intervention Pulse_Intervention Gradual_Intervention;
run;

/* -------------------------------------------------------
   PULSE MODEL: Test for Temporary Shock Effect
   ------------------------------------------------------- */
title4 "PULSE MODEL: Testing Temporary Effect";

proc arima data=plastik_pulse plots(unpack)=all;
    identify var=Volume_Sampah(1)
             crosscorr=(Pulse_Intervention(1))
             nlag=36;

    estimate q=(1,6) p=(12,24)
             input=(Pulse_Intervention)
             method=cls
             noconstant
             outest=pulse_params
             outstat=pulse_stats;

    forecast lead=24 id=Date interval=month out=pulse_forecast;
run;

/* -------------------------------------------------------
   COMBINED MODEL: Step + Pulse
   ------------------------------------------------------- */
title4 "COMBINED MODEL: Step + Pulse Intervention";

proc arima data=plastik_pulse plots(unpack)=all;
    identify var=Volume_Sampah(1)
             crosscorr=(Step_Intervention(1) Pulse_Intervention(1))
             nlag=36;

    estimate q=(1,6) p=(12,24)
             input=(Step_Intervention Pulse_Intervention)
             method=cls
             noconstant
             outest=combined_params
             outstat=combined_stats;

    forecast lead=24 id=Date interval=month out=combined_forecast;
run;

/* -------------------------------------------------------
   GRADUAL MODEL: Testing Gradual/Linear Effect
   ------------------------------------------------------- */
title4 "GRADUAL MODEL: Testing Linear Trend After Policy";

proc arima data=plastik_pulse plots(unpack)=all;
    identify var=Volume_Sampah(1)
             crosscorr=(Gradual_Intervention(1))
             nlag=36;

    estimate q=(1,6) p=(12,24)
             input=(Gradual_Intervention)
             method=cls
             noconstant
             outest=gradual_params
             outstat=gradual_stats;

    forecast lead=24 id=Date interval=month out=gradual_forecast;
run;

/* -------------------------------------------------------
   COMPARE ALL INTERVENTION TYPES
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```
   ------------------------------------------------------ */
title4 "INTERVENTION TYPE COMPARISON";

data intervention_comparison;
    length Model $50;

    set final_stats (in=a where=(_STAT_ in ('AIC', 'SBC')))
        pulse_stats (in=b where=(_STAT_ in ('AIC', 'SBC')))
        combined_stats (in=c where=(_STAT_ in ('AIC', 'SBC')))
        gradual_stats (in=d where=(_STAT_ in ('AIC', 'SBC')));

    if a then Model = "Step Intervention (Permanent)";
    else if b then Model = "Pulse Intervention (Temporary)";
    else if c then Model = "Combined (Step + Pulse)";
    else if d then Model = "Gradual (Linear Trend)";
run;

/* CRITICAL: Sort before transpose */
proc sort data=intervention_comparison;
    by Model;
run;

proc transpose data=intervention_comparison out=intervention_wide;
    by Model;
    id _STAT_;
    var _VALUE_;
run;

proc sort data=intervention_wide; by AIC; run;

title "BEST INTERVENTION TYPE (Lower AIC = Better)";
proc print data=intervention_wide noobs;
    format AIC SBC 12.2;
run;

/* Calculate intervention type differences */
data intervention_delta;
    set intervention_wide;

    if _N_ = 1 then do;
        Best_AIC = AIC;
        retain Best_AIC;
    end;

    Delta_AIC = AIC - Best_AIC;

    if Delta_AIC < 2 then Evidence = "Substantial support";
    else if Delta_AIC < 4 then Evidence = "Weak support";
    else if Delta_AIC < 7 then Evidence = "Little support";
    else Evidence = "No support";
run;

title "INTERVENTION TYPE COMPARISON WITH EVIDENCE";
proc print data=intervention_delta noobs;
    format AIC SBC Delta_AIC 12.2;
    var Model AIC SBC Delta_AIC Evidence;
run;

/* Extract Pulse Effect */
proc transpose data=pulse_params out=pulse_params_display;
    var _NUMERIC_;
    id _TYPE_;
run;

data pulse_effect;
    set pulse_params_display;
    where EST ne . and STD ne .;

    t_value = EST / STD;
    df = 120;
    p_value = 2 * (1 - probt(abs(t_value), df));
```

```
    CI_Lower = EST - tinv(0.975, df) * STD;
    CI_Upper = EST + tinv(0.975, df) * STD;

    if p_value < 0.001 then Sig = "***";
    else if p_value < 0.01 then Sig = "**";
    else if p_value < 0.05 then Sig = "*";
    else Sig = "ns";

    if p_value < 0.05 then
        Interpretation = "Significant TEMPORARY shock detected";
    else
        Interpretation = "NO significant temporary effect";
run;

title4 "PULSE INTERVENTION EFFECT";
proc print data=pulse_effect noobs;
    format EST STD CI_Lower CI_Upper 12.4 t_value 10.4 p_value pvalue6.4;
run;

/* Comparison Plot: Step vs Pulse vs Combined */
title "COMPARISON: Different Intervention Types";

ods graphics on / reset imagename="intervention_comparison" width=10in height=6in;

data forecast_combined;
    merge final_forecast (rename=(forecast=Step_Forecast) keep=Date Volume_Sampah forecast)
          pulse_forecast (rename=(forecast=Pulse_Forecast) keep=Date forecast)
          combined_forecast (rename=(forecast=Combined_Forecast) keep=Date forecast);
    by Date;
run;

proc sgplot data=forecast_combined;
    series x=Date y=Volume_Sampah /
        legendlabel="Actual"
        lineattrs=(color=black thickness=1);

    series x=Date y=Step_Forecast /
        legendlabel="Step Model"
        lineattrs=(color=blue thickness=2 pattern=solid);

    series x=Date y=Pulse_Forecast /
        legendlabel="Pulse Model"
        lineattrs=(color=green thickness=2 pattern=solid);

    series x=Date y=Combined_Forecast /
        legendlabel="Combined Model"
        lineattrs=(color=brown thickness=2 pattern=solid);

    refline '01MAR2022'd / axis=x
        lineattrs=(color=red pattern=shortdash thickness=2)
        label="Policy Implementation";

    refline '01SEP2025'd / axis=x
        lineattrs=(color=purple pattern=shortdash thickness=2)
        label="Forecast Start";

    xaxis label="Date" grid;
    yaxis label="Plastic Waste Volume (tons)" grid;

    title height=14pt "Comparison of Intervention Models";
    title2 height=11pt "Step (Permanent) vs Pulse (Temporary) vs Combined";
run;

ods graphics off;

/* ============================================================
   PART 8: FINAL MODEL DIAGNOSTICS
   ============================================================ */

title1 "==========================================================";
title2 "PART 8: FINAL MODEL DIAGNOSTICS";
title3 "==========================================================";
```

```
/* Residual white noise test */
title4 "8.1 Residual Autocorrelation - White Noise Check";
proc arima data=final_forecast plots=all;
    identify var=residual nlag=24;
run;

/* Normality test */
title4 "8.2 Normality Test of Residuals";
ods graphics on;
proc univariate data=final_forecast normal;
    var residual;
    histogram residual / normal;
    qqplot residual / normal(mu=est sigma=est);
run;
ods graphics off;

/* Parameter estimates with significance */
proc transpose data=final_params out=final_params_display;
    var _NUMERIC_;
    id _TYPE_;
run;

data final_params_test;
    set final_params_display;
    where EST ne . and STD ne .;

    t_value = EST / STD;
    df = 120;
    p_value = 2 * (1 - probt(abs(t_value), df));

    CI_Lower = EST - tinv(0.975, df) * STD;
    CI_Upper = EST + tinv(0.975, df) * STD;

    if p_value < 0.001 then Sig = "***";
    else if p_value < 0.01 then Sig = "**";
    else if p_value < 0.05 then Sig = "*";
    else Sig = "ns";

    keep _NAME_ EST STD t_value p_value CI_Lower CI_Upper Sig;
run;

title4 "8.3 FINAL MODEL: All Parameter Estimates";
proc print data=final_params_test noobs;
    format EST STD CI_Lower CI_Upper 12.6 t_value 10.4 p_value pvalue6.4;
run;

/* Calculate mean volume */
proc sql noprint;
    select mean(Volume_Sampah) into :mean_volume
    from plastik;
quit;

/* Extract intervention effect */
data intervention_effect;
    set final_params_test;
    where index(upcase(_NAME_), 'I1') > 0 or
          index(upcase(_NAME_), 'NUM') > 0 or
          index(upcase(_NAME_), 'INPUT') > 0;

    Pct_Change = (EST / &mean_volume) * 100;

    if p_value < 0.05 then do;
        if EST < 0 then
            Interpretation = "Significant DECREASE in waste volume";
        else
            Interpretation = "Significant INCREASE in waste volume";
    end;
    else do;
        Interpretation = "NO significant effect detected";
    end;
run;
```

```
title4 "8.4 INTERVENTION EFFECT: Statistical Results";
proc print data=intervention_effect noobs;
    format EST STD CI_Lower CI_Upper 12.4 p_value pvalue6.4 Pct_Change 8.2;
    var _NAME_ EST STD t_value p_value CI_Lower CI_Upper Pct_Change Sig Interpretation;
run;

/* ============================================================
   PART 9: MODEL ACCURACY METRICS
   ============================================================ */

title1 "==========================================================";
title2 "PART 9: MODEL ACCURACY";
title3 "==========================================================";

data accuracy;
    set final_forecast;
    where not missing(residual);

    AE = abs(residual);
    APE = abs(residual / Volume_Sampah) * 100;
    SE = residual**2;
    PE = (residual / Volume_Sampah) * 100;
run;

proc means data=accuracy noprint;
    var AE APE SE PE;
    output out=metrics
           mean=MAE MAPE MSE MPE
           std=StdAE StdAPE StdSE StdPE;
run;

data metrics_final;
    set metrics;
    RMSE = sqrt(MSE);

    if MAPE < 10 then Rating = "Excellent";
    else if MAPE < 20 then Rating = "Good";
    else if MAPE < 50 then Rating = "Reasonable";
    else Rating = "Poor";

    keep MAE MAPE RMSE MPE Rating;
run;

title4 "MODEL ACCURACY METRICS";
proc print data=metrics_final noobs;
    format MAE MAPE RMSE MPE 10.4;
run;

/* ============================================================
   PART 10: VISUALIZATION
   ============================================================ */

title1 "==========================================================";
title2 "PART 10: VISUALIZATION";
title3 "==========================================================";

ods graphics on / reset imagename="final_actual_vs_fitted" width=12in height=6in;

proc sgplot data=final_forecast;
    series x=Date y=Volume_Sampah /
        legendlabel="Actual Volume"
        lineattrs=(color=black thickness=2)
        markers markerattrs=(symbol=circlefilled size=6);

    series x=Date y=forecast /
        legendlabel="Fitted/Forecast Values"
        lineattrs=(color=blue thickness=2);

    band x=Date lower=L95 upper=U95 /
        legendlabel="95% Confidence Interval"
        transparency=0.3 fillattrs=(color=lightblue);
```

```
    refline '01MAR2022'd / axis=x
        lineattrs=(color=red pattern=shortdash thickness=3)
        label="Policy Implementation: March 2022"
        labelloc=inside labelattrs=(size=10pt weight=bold);

    refline '01SEP2025'd / axis=x
        lineattrs=(color=green pattern=shortdash thickness=3)
        label="Forecast Start: Sep 2025"
        labelloc=inside labelattrs=(size=10pt weight=bold color=green);

    xaxis label="Date (Monthly)" grid values=('01JAN2015'd to '01AUG2027'd by year);
    yaxis label="Plastic Waste Volume (tons)" grid;

    title height=14pt "FINAL MODEL: Actual vs Fitted/Forecast with Intervention Effect";
    title2 height=12pt "ARIMA Model + Step Intervention + 24-Month Forecast";
run;

ods graphics off;

/* Residuals plot */
ods graphics on / reset imagename="residuals_plot" width=10in height=5in;

proc sgplot data=final_forecast;
    scatter x=Date y=residual /
        markerattrs=(symbol=circlefilled size=5);

    refline 0 / axis=y lineattrs=(color=black pattern=solid);
    refline '01MAR2022'd / axis=x
        lineattrs=(color=red pattern=shortdash thickness=2);

    xaxis label="Date" grid;
    yaxis label="Residuals" grid;

    title "Residuals Over Time";
run;

ods graphics off;

/* ============================================================
   PART 10: BEST FORECAST VISUALIZATION
   ============================================================ */

title1 "==========================================================";
title2 "PART 10: COMPREHENSIVE FORECAST VISUALIZATION";
title3 "==========================================================";

/* ---------------------------------------------------------
   PLOT 1: ACTUAL vs FITTED (Main Result)
   Best plot showing model performance
   --------------------------------------------------------- */
title "MODEL PERFORMANCE: Actual vs Fitted Values";

ods graphics on / reset imagename="best_actual_vs_fitted" width=14in height=9in;

proc sgplot data=final_forecast;
    /* Actual data - prominent */
    series x=Date y=Volume_Sampah /
        lineattrs=(color=black thickness=3)
        markers
        markerattrs=(symbol=circlefilled size=8 color=black)
        legendlabel="Actual Volume"
        name="actual";

    /* Fitted values */
    series x=Date y=Forecast /
        lineattrs=(color=blue thickness=2 pattern=solid)
        legendlabel="Fitted Values (Model)"
        name="fitted";

    /* 95% Confidence interval */
    band x=Date lower=L95 upper=U95 /
```

```
        legendlabel="95% Confidence Interval"
        transparency=0.3
        fillattrs=(color=lightblue)
        name="ci";

    /* Policy implementation line */
    refline '01MAR2022'd / axis=x
        lineattrs=(color=red pattern=shortdash thickness=3)
        label="Policy Implementation (March 2022)"
        labelloc=inside
        labelattrs=(size=12pt weight=bold color=red);

    /* Axes */
    xaxis label="Time (Monthly)"
          grid
          gridattrs=(color=lightgray pattern=dot)
          values=('01JAN2015'd to '01DEC2025'd by year)
          valuesformat=year4.
          labelattrs=(size=12pt weight=bold);

    yaxis label="Plastic Waste Volume (tons)"
          grid
          gridattrs=(color=lightgray pattern=dot)
          labelattrs=(size=12pt weight=bold);

    /* Legend */
    keylegend "actual" "fitted" "ci" /
        location=inside
        position=topleft
        across=1
        title="Legend:"
        titleattrs=(size=11pt weight=bold)
        valueattrs=(size=10pt);

    /* Titles */
    title height=16pt bold "ARIMA Model Performance: Actual vs Fitted Values";
    title2 height=13pt "Plastic Waste Volume in Kota Palu (2015-2025)";
    title3 height=11pt italic "Blue Line = Model Prediction | Light Blue Band = 95% Confidence Interval";
run;

ods graphics off;

/* -------------------------------------------------------
   PLOT 2: PRE vs POST INTERVENTION COMPARISON
   Clear visualization of intervention effect
   ------------------------------------------------------- */

/* Create pre/post datasets */
data pre_post;
    set final_forecast;

    if Date < '01MAR2022'd then do;
        Period = "Pre-Policy";
        Period_Num = 1;
    end;
    else do;
        Period = "Post-Policy";
        Period_Num = 2;
    end;
run;

title "INTERVENTION EFFECT: Pre-Policy vs Post-Policy Comparison";

ods graphics on / reset imagename="intervention_effect_comparison" width=14in height=9in;

proc sgplot data=pre_post;
    /* Actual data with different colors for pre/post */
    scatter x=Date y=Volume_Sampah / group=Period
        markerattrs=(symbol=circlefilled size=8)
        name="actual";

    /* Fitted line */
```

```
    series x=Date y=Forecast /
        lineattrs=(color=darkblue thickness=2)
        legendlabel="Model Fit"
        name="fit";

    /* Intervention line */
    refline '01MAR2022'd / axis=x
        lineattrs=(color=red pattern=shortdash thickness=4)
        label="Policy Implementation"
        labelloc=inside
        labelattrs=(size=12pt weight=bold color=red);

    /* Shaded regions */
    refline '01MAR2022'd / axis=x
        lineattrs=(color=lightgray thickness=0)
        transparency=0.9;

    xaxis label="Time (Monthly)"
          grid
          values=('01JAN2015'd to '01DEC2025'd by year)
          valuesformat=year4.;

    yaxis label="Plastic Waste Volume (tons)"
          grid;

    keylegend "actual" "fit" /
        location=inside
        position=topright
        across=1;

    title height=16pt bold "Intervention Analysis: Visual Comparison";
    title2 height=13pt "Pre-Policy (Blue) vs Post-Policy (Orange) Period";
run;

ods graphics off;

/* -------------------------------------------------------
   PLOT 3: RESIDUALS ANALYSIS
   Model diagnostic plot
   ------------------------------------------------------- */
title "MODEL DIAGNOSTICS: Residuals Over Time";

ods graphics on / reset imagename="residuals_diagnostic" width=12in height=7in;

proc sgplot data=final_forecast;
    /* Scatter plot of residuals */
    scatter x=Date y=Residual /
        markerattrs=(symbol=circlefilled size=6 color=darkblue)
        transparency=0.3;

    /* Zero line */
    refline 0 / axis=y
        lineattrs=(color=black pattern=solid thickness=2)
        label="Zero Line";

    /* Policy line */
    refline '01MAR2022'd / axis=x
        lineattrs=(color=red pattern=dash thickness=2)
        label="Policy Implementation";

    /* Add LOESS smoother to see pattern */
    loess x=Date y=Residual /
        lineattrs=(color=red thickness=2)
        legendlabel="Trend (LOESS)";

    xaxis label="Time (Monthly)" grid;
    yaxis label="Residuals (tons)" grid;

    title height=14pt bold "Residual Analysis: Checking Model Assumptions";
    title2 height=11pt italic "Points should be randomly scattered around zero";
run;
```

```
ods graphics off;

/* ------------------------------------------------------
   PLOT 4: FORECAST ACCURACY BY YEAR
   Annual performance comparison
   ------------------------------------------------------ */

/* Calculate annual accuracy */
data annual_performance;
    set final_forecast;
    where not missing(Residual);

    Year = year(Date);

    /* Intervention period */
    if Date < '01MAR2022'd then Period = "Pre-Policy";
    else Period = "Post-Policy";

    /* Absolute error */
    Abs_Error = abs(Residual);
    Abs_Pct_Error = abs(Residual / Volume_Sampah) * 100;
run;

proc means data=annual_performance noprint;
    class Year;
    var Volume_Sampah Forecast Abs_Error Abs_Pct_Error;
    output out=annual_stats
           mean=Avg_Actual Avg_Forecast MAE MAPE
           std=Std_Actual Std_Forecast;
run;

data annual_stats;
    set annual_stats;
    where _TYPE_ = 1;
    Year_Label = put(Year, 4.);
run;

title "ANNUAL MODEL PERFORMANCE: Accuracy by Year";

ods graphics on / reset imagename="annual_accuracy" width=12in height=8in;

/* FIXED: TWO SEPARATE PLOTS */

/* Plot 1: Actual vs Forecast by Year */
proc sgplot data=annual_stats;
    series x=Year_Label y=Avg_Actual /
        lineattrs=(color=gray thickness=3)
        markers
        markerattrs=(symbol=circlefilled size=10 color=gray)
        legendlabel="Average Actual";

    series x=Year_Label y=Avg_Forecast /
        lineattrs=(color=blue thickness=3)
        markers
        markerattrs=(symbol=diamondfilled size=12 color=blue)
        legendlabel="Average Forecast";

    xaxis label="Year";
    yaxis label="Volume (tons)" grid;

    keylegend / location=inside position=topright;

    title height=14pt bold "Annual Performance: Actual vs Forecast";
run;

/* Plot 2: Error Trend */
proc sgplot data=annual_stats;
    series x=Year_Label y=MAE /
        lineattrs=(color=red thickness=3)
        markers
        markerattrs=(symbol=circlefilled size=10 color=red)
        legendlabel="Mean Absolute Error";
```

```
    xaxis label="Year";
    yaxis label="Mean Absolute Error (tons)" grid;

    title height=14pt bold "Annual Error Trend";
run;

ods graphics off;

/* -------------------------------------------------------
   PLOT 5: INTERVENTION IMPACT VISUALIZATION
   Before-After comparison with statistics
   ------------------------------------------------------- */

/* Calculate pre/post statistics */
proc means data=pre_post noprint;
    class Period;
    var Volume_Sampah Forecast;
    output out=period_stats
           mean=Mean_Actual Mean_Forecast
           std=Std_Actual Std_Forecast
           n=N;
run;

data period_stats;
    set period_stats;
    where _TYPE_ = 1;
run;

title "INTERVENTION IMPACT: Statistical Comparison";

ods graphics on / reset imagename="intervention_impact" width=10in height=8in;

/* FIXED: Combined bar chart */
data period_stats_combined;
    set period_stats;
    length Type $20;

    Type = "Actual";
    Value = Mean_Actual;
    output;

    Type = "Forecast";
    Value = Mean_Forecast;
    output;

    keep Period Type Value;
run;

proc sgplot data=period_stats_combined;
    vbar Period / response=Value
                  group=Type
                  groupdisplay=cluster
                  datalabel
                  datalabelattrs=(size=11pt weight=bold);

    xaxis label="Period" labelattrs=(size=13pt weight=bold);
    yaxis label="Average Volume (tons)" grid labelattrs=(size=13pt weight=bold);

    keylegend / location=inside position=topright;

    title height=15pt bold "Average Volume: Pre-Policy vs Post-Policy";
    title2 height=12pt "Actual vs Model Prediction Comparison";
run;

ods graphics off;
/* -------------------------------------------------------
   PLOT 6: MODEL FIT QUALITY
   Actual vs Fitted scatter plot
   ------------------------------------------------------- */
title "MODEL FIT QUALITY: Actual vs Fitted Scatter";
```

```
ods graphics on / reset imagename="fit_quality_scatter" width=10in height=10in;

proc sgplot data=final_forecast;
    /* Perfect fit line (45-degree) */
    lineparm x=0 y=0 slope=1 /
        lineattrs=(color=red pattern=dash thickness=2)
        legendlabel="Perfect Fit Line";

    /* Actual vs Fitted scatter */
    scatter x=Volume_Sampah y=Forecast /
        markerattrs=(symbol=circlefilled size=8 color=blue)
        transparency=0.4
        legendlabel="Data Points";

    /* Add regression line */
    reg x=Volume_Sampah y=Forecast /
        lineattrs=(color=darkblue thickness=2)
        legendlabel="Regression Line"
        clm
        alpha=0.05;

    xaxis label="Actual Volume (tons)" grid min=0;
    yaxis label="Fitted Volume (tons)" grid min=0;

    title height=14pt bold "Model Fit Quality: Actual vs Fitted";
    title2 height=11pt italic "Points closer to red line = better fit";
run;

ods graphics off;

/* -------------------------------------------------------
   SUMMARY TABLE WITH VISUALIZATION
   ------------------------------------------------------- */
title "COMPREHENSIVE MODEL SUMMARY";

/* Create summary statistics */
data model_summary;
    length Metric $40 Value $20;

    /* Get data from metrics_final and intervention_effect */
    set metrics_final(obs=1);

    Metric = "Mean Absolute Error (MAE)";
    Value = put(MAE, 10.2);
    output;

    Metric = "Mean Absolute % Error (MAPE)";
    Value = put(MAPE, 10.2) || "%";
    output;

    Metric = "Root Mean Square Error (RMSE)";
    Value = put(RMSE, 10.2);
    output;

    Metric = "Model Rating";
    Value = Rating;
    output;

    keep Metric Value;
run;

proc print data=model_summary noobs;
    var Metric Value;
    title4 "Model Accuracy Metrics";
run;

/* Print intervention effect */
proc print data=intervention_effect noobs;
    var _NAME_ EST p_value Pct_Change Interpretation;
    format EST 10.4 p_value pvalue6.4 Pct_Change 8.2;
    title4 "Intervention Effect Summary";
run;
```

```
/* ---------------------------------------------------------
   COMPLETION MESSAGE
   --------------------------------------------------------- */
data _null_;
    put "==========================================================";
    put "FORECAST VISUALIZATION COMPLETE!";
    put "==========================================================";
    put "Generated 6 comprehensive plots:";
    put "1. Actual vs Fitted (main result)";
    put "2. Pre vs Post intervention comparison";
    put "3. Residuals diagnostic plot";
    put "4. Annual accuracy by year";
    put "5. Intervention impact visualization";
    put "6. Model fit quality scatter";
    put " ";
    put "All plots show in-sample model performance (2015-2025)";
    put "==========================================================";
run;

/* ============================================================
   PART 10B: CREATE 24-MONTH FORECAST DATASETS
   ============================================================ */

title1 "==========================================================";
title2 "PART 10B: 24-MONTH FORECAST DATA PREPARATION";
title3 "==========================================================";

/* Create labeled forecast dataset */
data forecast_24month_labeled;
    set final_forecast;

    Year = year(Date);
    Month = month(Date);
    YearMonth = put(Date, yymms7.);

    /* Mark historical vs forecast period */
    if Date <= '01AUG2025'd then Period = "Historical";
    else Period = "Forecast";

    /* Calculate error range */
    Error_Range = U95 - L95;

    /* Keep only necessary variables */
    keep Date Volume_Sampah Forecast STD L95 U95 Residual
         Year Month YearMonth Period Error_Range;
run;

/* Verify the dataset */
title4 "Forecast Dataset Verification";
proc print data=forecast_24month_labeled(obs=10);
    var Date Period Forecast L95 U95;
    format Date yymmdd10. Forecast L95 U95 10.2;
run;

/* Create forecast-only dataset (Sep 2025 - Aug 2027) */
data forecast_only;
    set forecast_24month_labeled;
    where Date > '01AUG2025'd;
run;

/* Verify forecast-only */
title4 "Forecast-Only Data (Sep 2025 onwards)";
proc print data=forecast_only(obs=24);
    var Date YearMonth Forecast L95 U95 Error_Range;
    format Forecast L95 U95 Error_Range 10.2;
run;

/* Create annual summary for forecast period */
proc means data=forecast_only noprint;
    class Year;
    var Forecast L95 U95;
```

```
    output out=annual_summary
           mean=Avg_Forecast Avg_L95 Avg_U95
           std=Std_Forecast
           n=N_Months;
run;

/* Clean annual summary */
data annual_summary;
    set annual_summary;
    where _TYPE_ = 1 and Year ne .;
    Year_Label = put(Year, 4.);

    /* Calculate forecast range */
    CI_Width = Avg_U95 - Avg_L95;

    keep Year Year_Label N_Months Avg_Forecast Avg_L95 Avg_U95
         Std_Forecast CI_Width;
run;

/* Print annual summary */
title4 "Annual Forecast Summary";
proc print data=annual_summary noobs;
    format Avg_Forecast Avg_L95 Avg_U95 Std_Forecast CI_Width 10.2;
run;

/* Check dataset counts */
data _null_;
    put "=========================================================";
    put "FORECAST DATASETS CREATED SUCCESSFULLY";
    put "=========================================================";

    /* Get counts */
    dsid1 = open('forecast_24month_labeled');
    nobs1 = attrn(dsid1, 'NOBS');
    rc1 = close(dsid1);

    dsid2 = open('forecast_only');
    nobs2 = attrn(dsid2, 'NOBS');
    rc2 = close(dsid2);

    dsid3 = open('annual_summary');
    nobs3 = attrn(dsid3, 'NOBS');
    rc3 = close(dsid3);

    put "1. forecast_24month_labeled: " nobs1 "observations";
    put "2. forecast_only: " nobs2 "observations";
    put "3. annual_summary: " nobs3 "observations";
    put "=========================================================";
run;

/* ============================================================
   PART 11: EXPORT RESULTS TO CSV
   ============================================================ */

title1 "=========================================================";
title2 "PART 11: EXPORT RESULTS";
title3 "=========================================================";

/* Define output path */
%let output_path = /home/u63802153/sasuser.v94/Hasil SAS;

proc export data=comparison_wide
    outfile="&output_path./Model_Comparison.csv"
    dbms=csv replace;
run;

proc export data=final_forecast
    outfile="&output_path./Final_Forecast_Results.csv"
    dbms=csv replace;
run;

proc export data=final_params_test
```

```
    outfile="&output_path./Final_Parameters.csv"
    dbms=csv replace;
run;

proc export data=intervention_effect
    outfile="&output_path./Intervention_Effect.csv"
    dbms=csv replace;
run;

proc export data=metrics_final
    outfile="&output_path./Model_Accuracy.csv"
    dbms=csv replace;
run;

proc export data=intervention_wide
    outfile="&output_path./Intervention_Type_Comparison.csv"
    dbms=csv replace;
run;

proc export data=forecast_24month_labeled
    outfile="&output_path./Forecast_24Month_Sep2025_Aug2027.csv"
    dbms=csv replace;
run;

proc export data=annual_summary
    outfile="&output_path./Annual_Forecast_Summary.csv"
    dbms=csv replace;
run;

proc export data=forecast_only
    outfile="&output_path./Monthly_Forecast_24Months.csv"
    dbms=csv replace;
run;

proc export data=plastik_yearly
    outfile="&output_path./Data_Yearly_BoxPlot.csv"
    dbms=csv replace;
run;

/* ============================================================
   PART 12: FINAL SUMMARY REPORT
   ============================================================ */

title1 "====================================================================";
title2 "            FINAL ANALYSIS SUMMARY REPORT                           ";
title3 "====================================================================";

proc print data=comparison_delta(obs=4) noobs;
    title4 "TOP 4 ARIMA MODELS BY AIC";
    format AIC SBC Delta_AIC Delta_SBC 12.2;
    var Model AIC SBC Delta_AIC Evidence;
run;

proc print data=intervention_delta noobs;
    title4 "INTERVENTION TYPE COMPARISON";
    format AIC SBC Delta_AIC 12.2;
    var Model AIC SBC Delta_AIC Evidence;
run;

proc print data=final_params_test noobs;
    title4 "FINAL MODEL: All Parameters with Significance";
    format EST STD CI_Lower CI_Upper 12.6 t_value 10.4 p_value pvalue6.4;
run;

proc print data=intervention_effect noobs;
    title4 "INTERVENTION EFFECT: Key Findings";
    format EST STD CI_Lower CI_Upper 12.4 p_value pvalue6.4 Pct_Change 8.2;
run;

proc print data=metrics_final noobs;
    title4 "MODEL ACCURACY";
    format MAE MAPE RMSE MPE 10.4;
```

```
run;

/* Data Summary */
title4 "DATA SUMMARY";
proc means data=plastik n mean std min max;
    var Volume_Sampah;
run;

proc freq data=plastik;
    tables Intervention;
    title4 "Intervention Period Distribution";
run;

/* 24-Month Forecast Summary */
title4 "24-MONTH FORECAST SUMMARY (Sep 2025 - Aug 2027)";
proc means data=forecast_only n mean std min max;
    var Forecast L95 U95;
run;

/* ============================================================
   END OF SCRIPT

   SUMMARY OF CORRECTIONS:
   1. Fixed typo in PART 7 comment (//* -> /*)
   2. Consistent model naming throughout:
      - Model 1: ARIMA(0,1,1)(1,0,0)12
      - Model 2: ARIMA(0,1,1)(2,0,0)12
```